

Xuanyao Fong

其他

基本信息

姓名: Xuanyao Fong

所属院校: 新加坡国立大学

地区: Singapore

海外经历: 有

学科领域

应用物理学

微电子科学与工程

电子与计算机工程

集成电路设计与集成系统

电子科学与技术

人工智能

研究领域

磁学与自旋电子学

微电子器件与制造

低功耗设计与能耗优化

深度学习与神经网络

纳米电子与先进工艺

存储器设计与架构

微纳电子与纳米技术

半导体器件物理与工艺

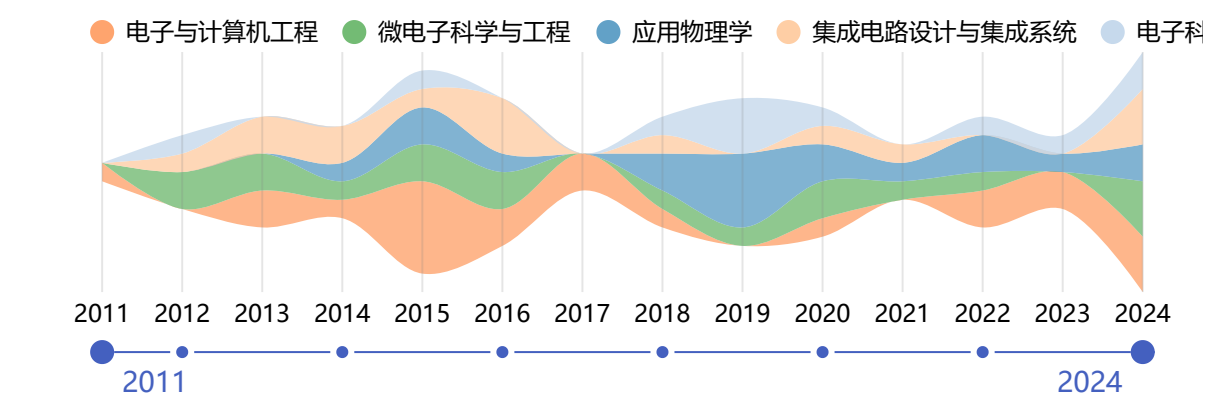
个人简介

Xuanyao Fong is affiliated with the National University of Singapore. His research focuses on devices-to-systems co-design methodologies, compact modeling, electronic device physics and technology, spintronics, and VLSI. Notable contributions include work on spin-transfer torque devices for logic and memory, spin-transfer torque memories, and multilevel spin-orbit torque MRAMs. His research has been widely cited in leading journals and conferences. Email: Verified email at nus.edu.sg.

工作经历

National University of Singapore

近年发表的论文



Bimodal alteration of cognitive accuracy for spintronic artificial neural networks

NANOSCALE HORIZONS

Kumar, Anuj; Das, Debasis; Lin, Dennis J. X.; Huang, Lisen; Yap, Sherry L. K.; Tan, Hang Khume; Lim, Royston J. J.; Tan, Hui Ru; Toh, Yeow Teck; Lim, Sze Ter; Fong, Xuanyao; Ho, Pin

被引用次数: 0 2024

Multistate Compound Magnetic Tunnel Junction Synapses for Digital Recognition

ACS APPLIED MATERIALS & INTERFACES

Kumar, Anuj; Lin, Dennis J. X.; Das, Debasis; Huang, Lisen; Yap, Sherry L. K.; Tan, Hui Ru; Tan, Hang Khume; Lim, Royston J. J.; Toh, Yeow Teck; Chen, Shaohai; Lim, Sze Ter; Fong, Xuanyao; Ho, Pin

被引用次数: 3 2024

Energy-efficient superparamagnetic Ising machine and its application to traveling salesman problems

NATURE COMMUNICATIONS

Si, Jia; Yang, Shuhan; Cen, Yunuo; Chen, Jiaer; Huang, Yingna; Yao, Zhaoyang; Kim, Dong-Jun; Cai, Kaiming; Yoo, Jerald; Fong, Xuanyao; Yang, Hyunsoo

被引用次数: 7 2024

Modeling of Ferroelectric Thin Film Transistors with Amorphous Oxide Semiconductor Channel

8TH IEEE ELECTRON DEVICES TECHNOLOGY & MANUFACTURING CONFERENCE, EDTM 2024

Zhang, Wei; Wang, Jianze; Sun, Chen; Wu, Zhen; Gong, Xiao; Fong, Xuanyao

被引用次数: 0 2024

Energy-Efficient Ising Machines Using Capacitance-Coupled Latches for MaxCut Solving

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Wang, Yimin; Fong, Xuanyao

被引用次数: 0 2024

A Failure Analysis Framework For Ferroelectric Field-Effect Transistor Memory

2024 IEEE INTERNATIONAL CONFERENCE ON IC DESIGN AND TECHNOLOGY, IICIDT 2024

Wang, Jianze; Zhang, Wei; Fong, Xuanyao

被引用次数: 0

2024

3D Phase Field Simulation and Variability Analysis of Polycrystalline Ferroelectric Memories

2024 IEEE INTERNATIONAL CONFERENCE ON IC DESIGN AND TECHNOLOGY, ICICDT 2024

Zhang, Wei; Xie, Yuhang; Wang, Jianze; Fong, Xuanyao

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2024

Transposable Memory Based on the Ferroelectric Field-Effect Transistor

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Wang, Jianze; Zhang, Wei; Wu, Zhen; Wang, Yimin; Jiao, Leming; Wang, Xiaolin; Gong, Xiao; Fong, Xuanyao

被引用次数: 1

2024

Design Framework for Ising Machines with Bistable Latch -Based Spins and All -to -All Resistive Coupling

2024 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS, ISCAS 2024

Wang, Yimin; Cen, Yunuo; Fong, Xuanyao

被引用次数: 0

2024

Spintronic devices for high-density memory and neuromorphic computing - A review MATERIALS TODAY

Chen, Bingjin; Zeng, Minggang; Khoo, Khoong Hong; Das, Debasis; Fong, Xuanyao; Fukami, Shunsuke; Li, Sai; Zhao, Weisheng; Parkin, Stuart S. P.; Piramanayagam, S. N.; Ter Lim, Sze

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2023

Benchmarking DNN Mapping Methods for the in-Memory Computing Accelerators

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Wang, Yimin; Fong, Xuanyao

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2023

Attack Resilient True Random Number Generators Using Ferroelectric-Enhanced Stochasticity in 2D Transistor

SMALL

Chien, Yu-Chieh; Xiang, Heng; Wang, Jianze; Shi, Yufei; Fong, Xuanyao; Ang, Kah-Wee

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Bilayer-Skyrmion-Based Design of Neuron and Synapse for Spiking Neural Network

PHYSICAL REVIEW APPLIED

Das, Debasis; Cen, Yunuo; Wang, Jianze; Fong, Xuanyao

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2023

A tree search algorithm towards solving Ising formulated combinatorial optimization problems

SCIENTIFIC REPORTS

Cen, Yunuo; Das, Debasis; Fong, Xuanyao

被引用次数: 2

2022

Self-Reset Schemes for Magnetic Domain Wall-Based Neuron

IEEE JOURNAL ON EXPLORATORY SOLID-STATE COMPUTATIONAL DEVICES AND CIRCUITS

Das, Debasis; Fong, Xuanyao

被引用次数: 1

2022

Wafer-Scale 2D Hafnium Diselenide Based Memristor Crossbar Array for Energy-Efficient Neural Network Hardware

ADVANCED MATERIALS

Li, Sifan; Pam, Mei-Er; Li, Yesheng; Chen, Li; Chien, Yu-Chieh; Fong, Xuanyao; Chi, Dongzhi; Ang, Kah-Wee

被引用次数: 125

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Spintronic Integrate-Fire-Reset Neuron with Stochasticity for Neuromorphic Computing

NANO LETTERS

Yang, Qu; Mishra, Rahul; Cen, Yunuo; Shi, Guoyi; Sharma, Raghav; Fong, Xuanyao; Yang, Hyunsoo

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An FPGA-Based Co-Processor for Spiking Neural Networks with On-Chip STDP-Based Learning

2022 IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS (ISCAS 22)

Nguyen, Thao N. N.; Veeravalli, Bharadwaj; Fong, Xuanyao

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Memtransistor-Based Ternary Content-Addressable Memories: Design and Evaluation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Ly, Denys R. B.; Fong, Xuanyao

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Impact of Trap Profile on the Characteristics of 2-D MoS₂ Memtransistors: A Simulation Study

IEEE TRANSACTIONS ON ELECTRON DEVICES

Zhang, Panpan; Feng, Xuewei; Fong, Xuanyao

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2022

A Fokker-Planck Approach for Modeling the Stochastic Phenomena in Magnetic and Resistive Random Access Memory Devices

IEEE TRANSACTIONS ON ELECTRON DEVICES

Das, Debasis; Fong, Xuanyao

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2021

Self-Selective Multi-Terminal Memtransistor Crossbar Array for In-Memory Computing

ACS NANO

Feng, Xuewei; Li, Sifan; Wong, Swee Liang; Tong, Shiwun; Chen, Li; Zhang, Panpan; Wang, Lingfei; Fong, Xuanyao; Chi, Dongzhi; Ang, Kah-Wee

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2021

Electron-beam-irradiated rhenium disulfide memristors with low variability for neuromorphic computing

NPJ 2D MATERIALS AND APPLICATIONS

Li, Sifan; Li, Bochang; Feng, Xuewei; Chen, Li; Li, Yesheng; Huang, Li; Fong, Xuanyao; Ang, Kah-Wee

被引用次数: 41

2021

Physical Insights Into the Mobility Enhancement in Amorphous InGaZnO Thin-Film Transistor by SiO₂ Passivation Layer

IEEE TRANSACTIONS ON ELECTRON DEVICES

Zhang, Panpan; Samanta, Subhranu; Fong, Xuanyao

被引用次数: 4

2020

SIMBA: A Skyrmionic In-Memory Binary Neural Network Accelerator

IEEE TRANSACTIONS ON MAGNETICS

Miriyala, Venkata Pavan Kumar; Vishwanath, Kale Rahul; Fong, Xuanyao

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2020

Voltage-Controlled Spintronic Stochastic Neuron for Restricted Boltzmann Machine With Weight Sparsity

IEEE ELECTRON DEVICE LETTERS

Deng, Jiefang; Miriyala, Venkata Pavan Kumar; Zhu, Zhifeng; Fong, Xuanyao; Liang, Gengchiao

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2020

Electrical Generation and Detection of Terahertz Signal Based on Spin-Wave Emission From Ferrimagnets

PHYSICAL REVIEW APPLIED

Zhu, Zhifeng; Cai, Kaiming; Deng, Jiefang; Miriyala, Venkata Pavan Kumar; Yang, Hyunsoo; Fong, Xuanyao; Liang, Gengchiao

被引用次数: 10

2020

Low Subthreshold Swing and High Mobility Amorphous Indium-Gallium-Zinc-Oxide Thin-Film Transistor With Thin HfO₂ Gate Dielectric and Excellent Uniformity

IEEE ELECTRON DEVICE LETTERS

Samanta, Subhranu; Chand, Umesh; Xu, Shengqiang; Han, Kaizhen; Wu, Ying; Wang, Chengkuan; Kumar, Annie; Velluri, Hasita; Li, Yida; Fong, Xuanyao; Thean, Aaron Voon-Yew; Gong, Xiao

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2020

Transition from trap-mediated to band-like transport in polycrystalline monolayer molybdenum disulfide memtransistors

APPLIED PHYSICS LETTERS

Zhang, Panpan; Wang, Lin; Ang, Kah-Wee; Fong, Xuanyao

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2020

Voltage-input spintronic oscillator based on competing effect for extended oscillation regions

JOURNAL OF APPLIED PHYSICS

Zhu, Zhifeng; Deng, Jiefang; Fong, Xuanyao; Liang, Gengchiao

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Influence of Size and Shape on the Performance of VCMA-Based MTJs

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Miriyala, Venkata Pavan Kumar; Fong, Xuanyao; Liang, Gengchiao

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2019

Spin-wave mediated interactions for majority computation using Skyrmions and spin-torque nano-oscillators

JOURNAL OF MAGNETISM AND MAGNETIC MATERIALS

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2019

Bipolar resistive switching and synaptic characteristics modulation at sub- μ A current level using novel Ni/SiO_x/W cross-point structure

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2019

Domain Wall Motion-Based Dual-Threshold Activation Unit for Low-Power Classification of Non-Linearly Separable Functions

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Deb, Suman; Vatwani, Tarun; Chattopadhyay, Anupam; Basu, Arindam; Fong, Xuanyao

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2018

Configurable multifunctional integrated circuits based on carbon nanotube dual-material gate devices

NANOSCALE

Xiang, Li; Wang, Yuwei; Zhang, Panpan; Fong, Xuanyao; Wei, Xianlong; Hu, Youfan

被引用次数: 9

2018

Damping-like spin-orbit-torque-induced magnetization dynamics in ferrimagnets based on Landau-Lifshitz-Bloch equation

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Zhu, Zhifeng; Fong, Xuanyao; Liang, Gengchiao

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Electric-field-induced three-terminal pMTJ switching in the absence of an external magnetic field

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2018

Theoretical proposal for determining angular momentum compensation in ferrimagnets

PHYSICAL REVIEW B

Zhu, Zhifeng; Fong, Xuanyao; Liang, Gengchiao

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2018

Fast and Disturb-Free Nonvolatile Flip-Flop Using Complementary Polarizer MTJ

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Seo, Yeongkyo; Fong, Xuanyao; Roy, Kaushik

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2017

Yield, Area, and Energy Optimization in STT-MRAMs Using Failure-Aware ECC

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Pajouhi, Zoha; Fong, Xuanyao; Raghunathan, Anand; Roy, Kaushik

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2017

Spin-Transfer Torque Memories: Devices, Circuits, and Systems

PROCEEDINGS OF THE IEEE

Fong, Xuanyao; Kim, Yusung; Venkatesan, Rangharajan; Choday, Sri Harsha; Raghunathan, Anand; Roy, Kaushik

被引用次数: 155

2016

High Performance and Energy-Efficient On-Chip Cache Using Dual Port (1R/1W) Spin-Orbit Torque MRAM

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Seo, Yeongkyo; Kwon, Kon-Woo; Fong, Xuanyao; Roy, Kaushik

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Comprehensive Scaling Analysis of Current Induced Switching in Magnetic Memories Based on In-Plane and Perpendicular Anisotropies

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Jaiswal, Akhilesh; Fong, Xuanyao; Roy, Kaushik

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2016

Embedding Read-Only Memory in Spin-Transfer Torque MRAM-Based On-Chip Caches

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Fong, Xuanyao; Venkatesan, Rangharajan; Lee, Dongsoo; Raghunathan, Anand; Roy, Kaushik

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Spin-Transfer Torque Devices for Logic and Memory: Prospects and Perspectives

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Fong, Xuanyao; Kim, Yusung; Yogendra, Karthik; Fan, Deliang; Sengupta, Abhronil; Raghunathan, Anand; Roy, Kaushik

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Modeling and Evaluation of Topological Insulator/Ferromagnet Heterostructure-Based Memory

IEEE TRANSACTIONS ON ELECTRON DEVICES

Reza, Ahmed Kamal; Fong, Xuanyao; Al Azim, Zubair; Roy, Kaushik

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2016

Spin-Hall Magnetic Random-Access Memory With Dual Read/Write Ports for On-Chip Caches

IEEE MAGNETICS LETTERS

Seo, Yeongkyo; Fong, Xuanyao; Kwon, Kon-Woo; Roy, Kaushik

被引用次数: 25

2015

Spin-Orbit-Torque-Based Spin-Dice: A True Random-Number Generator

IEEE MAGNETICS LETTERS

Kim, Yusung; Fong, Xuanyao; Roy, Kaushik

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2015

High-Density and Robust STT-MRAM Array Through Device/Circuit/Architecture Interactions

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Kwon, Kon-Woo; Fong, Xuanyao; Wijesinghe, Parami; Panda, Priyadarshini; Roy, Kaushik

被引用次数: 30

2015

Highly Reliable Spin-Transfer Torque Magnetic RAM-Based Physical Unclonable Function With Multi-Response-Bits Per Cell

IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY

Zhang, Le; Fong, Xuanyao; Chang, Chip-Hong; Kong, Zhi Hui; Roy, Kaushik

被引用次数: 33

2015

Multilevel Spin-Orbit Torque MRAMs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Kim, Yusung; Fong, Xuanyao; Kwon, Kon-Woo; Chen, Mei-Chin; Roy, Kaushik

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2015

Domain Wall Coupling-Based STT-MRAM for On-Chip Cache Applications

IEEE TRANSACTIONS ON ELECTRON DEVICES

Seo, Yeongkyo; Fong, Xuanyao; Roy, Kaushik

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2015

Optimizing Emerging Nonvolatile Memories for Dual-Mode Applications: Data Storage and Key Generator

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Zhang, Le; Fong, Xuanyao; Chang, Chip-Hong; Kong, Zhi Hui; Roy, Kaushik

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2015

Spin-orbit torque induced spike-timing dependent plasticity

APPLIED PHYSICS LETTERS

Sengupta, Abhronil; Al Azim, Zubair; Fong, Xuanyao; Roy, Kaushik

被引用次数: 94

2015

Exploring Spin Transfer Torque Devices for Unconventional Computing

IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS

Roy, Kaushik; Fan, Deliang; Fong, Xuanyao; Kim, Yusung; Sharad, Mrigank; Paul, Somnath; Chatterjee, Subho; Bhunia, Swarup; Mukhopadhyay, Saibal

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2015

Failure Mitigation Techniques for 1T-1MTJ Spin-Transfer Torque MRAM Bit-cells

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Fong, Xuanyao; Kim, Yusung; Choday, Sri Harsha; Roy, Kaushik

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2014

Non-Volatile Complementary Polarizer Spin-Transfer Torque On-Chip Caches: A Device/Circuit/Systems Perspective

IEEE TRANSACTIONS ON MAGNETICS

Fong, Xuanyao; Venkatesan, Rangharajan; Raghunathan, Anand; Roy, Kaushik

被引用次数: 14

2014

Laser Induced Magnetization Reversal for Detection in Optical Interconnects

IEEE ELECTRON DEVICE LETTERS

Al Azim, Zubair; Fong, Xuanyao; Ostler, Thomas; Chantrell, Roy; Roy, Kaushik

被引用次数: 5

2014

SHE-NVFF: Spin Hall Effect-Based Nonvolatile Flip-Flop for Power Gating Architecture

IEEE ELECTRON DEVICE LETTERS

Kwon, Kon-Woo; Choday, Sri Harsha; Kim, Yusung; Fong, Xuanyao; Park, Sang Phill; Roy, Kaushik

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2014

R-MRAM: A ROM-Embedded STT MRAM Cache

IEEE ELECTRON DEVICE LETTERS

Lee, Dongsoo; Fong, Xuanyao; Roy, Kaushik

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2013

Dual Pillar Spin-Transfer Torque MRAMs for Low Power Applications

ACM JOURNAL ON EMERGING TECHNOLOGIES IN COMPUTING SYSTEMS

Mojumder, Niladri N.; Fong, Xuanyao; Augustine, Charles; Gupta, Sumeet K.; Choday, Harsha; Roy, Kaushik

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2013

Complimentary Polarizers STT-MRAM (CPSTT) for On-Chip Caches

IEEE ELECTRON DEVICE LETTERS

Fong, Xuanyao; Roy, Kaushik

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Complimentary Polarizers STT-MRAM (CPSTT) for On-Chip Caches (vol 34, pg 232, 2013)

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2013

Spin-Transfer Torque MRAMs for Low Power Memories: Perspective and Prospective

IEEE SENSORS JOURNAL

Augustine, Charles; Mojumder, Niladri Narayan; Fong, Xuanyao; Choday, Harsha; Park, Sang Phill; Roy, Kaushik

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2012

Bit-Cell Level Optimization for Non-volatile Memories Using Magnetic Tunnel Junctions and Spin-Transfer Torque Switching

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Fong, Xuanyao; Choday, Sri Harsha; Roy, Kaushik

被引用次数: 71

2012

Ultra-Low Power Nanomagnet-Based Computing: A System-Level Perspective

IEEE TRANSACTIONS ON NANOTECHNOLOGY

Augustine, Charles; Fong, Xuanyao; Behin-Aein, Behtash; Roy, Kaushik

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2011

Highly Reliable Spin-Transfer Torque Magnetic RAM-Based Physical Unclonable Function With Multi-Response-Bits Per Cell

TIFS

Le Zhang; Xuanyao Fong; Chip-Hong Chang; Zhi Hui Kong; Kaushik Roy

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Spin-Transfer Torque Devices for Logic and Memory: Prospects and Perspectives

TCAD

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TCAD

Le Zhang; Xuanyao Fong; Chip-Hong Chang; Zhi Hui Kong; Kaushik Roy

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Spin-Transfer Torque Memories: Devices, Circuits, and Systems

Proc.IEEE

Xuanyao Fong; Yusung Kim; Rangharajan Venkatesan; Sri Harsha Choday; Anand Raghunathan; Kaushik Roy

被引用次数: 0

学术合作



Gengchiau Liang

台湾阳明交通大学

合作次数: 9

Jiefang Deng

新加坡国立大学

合作次数: 5

Venkata Pavan Kumar Miriyala

新加坡国立大学

合作次数: 5

Yunuo Cen

新加坡国立大学

合作次数: 5

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新加坡国立大学

合作次数: 4

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